Name

Mathematics Advanced Year 12

Sequences and Series

Book 1

Version: 261021

Feedback:
https://MrDingMaths.com

Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-12-03 uses arithmetic and geometric sequences and series to model and solve problems

Sequences and series

- Define a sequence as an ordered list of objects
- Use the notation a_n, where n is a positive integer, to represent the nth term of a sequence
- Distinguish between a finite sequence $a_1, a_2, \ldots, a_m$ that terminates at its mth term, for some whole number $m \geq 1$, and an infinite sequence $a_1, a_2, \ldots$ that never terminates
- Define the nth partial sum S_n of a sequence $a_1, a_2, \ldots$ to be the sum of the first n terms of the sequence: $S_n = a_1 + a_2 + \cdots + a_n$, for all whole numbers $n \geq 1$
- Define a series formally as the sum of the terms of an infinite sequence and use the notation $a_1 + a_2 + a_3 + \cdots$ for the series corresponding to the sequence $a_1, a_2, a_3 \ldots$
- Use summation notation to represent the sum of terms a_i to a_j of a sequence where $j > i$, $\sum_{k=i}^{j} a_k = a_i + a_{i+1} + a_{i+2} + \ \ldots \ + a_{j-1} + a_j$

Sequences

ⓘ Sequences

A sequence is a collection of terms arranged in a specific order, where each term is obtained according to a rule.

Some examples of sequences:

2, 4, 6, 8, 10 1, 4, 9, 16, 25 1, 2, 4, 8, 16 1, 1, 2, 3, 5, 8

ⓘ Terms of a Sequence

We express terms of a sequence as $a_1, a_2, a_3, \ldots a_n$ where:

a_1 is the 1st term

a_2 is the 2nd term

a_n is the nth term

⚠ n must be a positive integer $n \geq 1$

ⓘ Recursive Rule

A **recursive rule**, or **term-to-term rule**, describes how to find the next term in a sequence by using the current term. It shows the operation(s) you perform on one term to get the following term.

E.g. $a_n = a_{n-1} + 3$ means: "to get the next term, add 3 to the previous term"

ⓘ n^{th} Term Formula

The nth **term formula,** or **position-to-term rule**, directly calculates any term using its position number in the sequence.

This is more powerful than a term-to-term rule because you can instantly find any term without working through the entire sequence.

E.g., $a_n = 3n + 5$ means: "to get the nth term, multiply by 3 and add 5"

Interpret rules of sequences.

A sequence has a rule: $a_n = 7n - 3$.

Write down the first five terms of the sequence.

Position n	1	2	3	4	5
Term a_n					

Describe the term-to-term rule in words: ……

Describe the term-to-term rule using the notation: $a_n =$ ……

A sequence has a rule: $a_n = a_{n-1} + 10$ and $a_1 = 14$.

Write down the first five terms of the sequence.

Position n	1	2	3	4	5
Term a_n					

Find a formula for the nth term $a_n =$ ……

a A sequence has a rule: $a_n = 2^n - 1$.
Write down the first five terms of the sequence.

Position n	1	2	3	4	5
Term a_n					

Describe the term-to-term rule in words: ……

Describe the term-to-term rule using the notation: $a_n =$ ……

b A sequence has a rule: $a_n = a_{n-1} \times 2$, where $a_1 = \frac{1}{4}$.
Write down the first five terms of the sequence.

Position n	1	2	3	4	5
Term a_n					

Find a formula for the nth term $a_n =$ ……

ⓘ Using the nth Term Formula

Many problems can be solved by forming an equation using the nth term formula.

Example Solving problems using the nth term formula

Find whether 400 is a term of the sequence $a_n = 7n + 20$.

$$400 = 7n + 20$$
$$380 = 7n$$
$$n = 54.2867 \ldots$$

$\therefore$ no, as n is not a whole number.

Find how many negative terms are in $a_n = 12n - 100$.

$$12n - 100 < 0$$
$$12n < 100$$
$$n < 8.3333$$

There are 8 negative terms.

Practice

Find whether 300 is a term of the sequence $a_n = 7n + 20$.

yes

Find whether 485 is a term of the sequence $a_n = 2n - 16$

yes

Find the *position* of the first positive term of $a_n = 7n - 60$.

9

Find the first positive term of $a_n = 7n - 60$.

3

1 Write down the next four terms of each sequence.

a 6, 16, 26, ...

36, 46, 56, 66

b 3, 6, 12, ...

24, 48, 96, 192

c 38, 34, 30, ...

26, 22, 18, 14

d 24, 12, 6, ...

$3, 1\frac{1}{2}, \frac{3}{4}, \frac{3}{8}$

e -1, 1, -1, ...

$1, -1, 1, -1$

f 1, 4, 9, ...

16, 25, 36, 49

g $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$

$\frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$

h 16, -8, 4, ...

$-2, 1, -\frac{1}{2}, \frac{1}{4}$

2 Find the first four terms of the sequence whose n^{th} term a_n is:

a $a_n = 5n - 2$

3, 8, 13, 18

b $a_n = 5^n$

5, 25, 125, 625

c $a_n = 6 - 2n$

4, 2, 0, −2

d $a_n = 7 \times 10^n$

70, 700, 7000, 70000

e $a_n = n^3$

1, 8, 27, 64

f $a_n = n(n + 1)$

2, 6, 12, 20

g $a_n = (-1)^n$

$-1, 1, -1, 1$

h $a_n = (-3)^n$

$-3, 9, -27, 81$

3 Write down the first four terms of each sequence described below.

a First term is 11, every term after is 50 more than previous

11, 61, 111, 161

b First term is 15, every term after is 3 less than previous

15, 12, 9, 6

c First term is 5, every term after is twice the previous

5, 10, 20, 40

d First term is -100, every term after is one fifth of previous

$-100, -20, -4, -\frac{4}{5}$

4 Consider the sequence: 7, 12, 17, 22, ...

a Write the first 12 terms.

7, 12, 17, 22, 27, 32, 37, 42, 47, 52, 57, 62

b How many terms are less than 30?

5

c How many terms lie between 20 and 40?

4

d What is the 10th term?

52

e What number term is 37?

7th term

f Is 87 a term in the sequence?

Yes; 17th term

g Is 201 a term in the sequence?

No; they all end in 2 or 7

5 For each sequence, write out the first five terms.
Then describe how each term is obtained from the previous term.

a $a_n = 12 + n$

13, 14, 15, 16, 17. Add 1.

b $a_n = 4 + 5n$

9, 14, 19, 24, 29. Add 5.

c $a_n = 15 - 5n$

10, 5, 0, -5, -10. Subtract 5.

d $a_n = 3 \times 2^n$

6, 12, 24, 48, 96. Multiply by 2.

e $a_n = 7 \times (-1)^n$

-7, 7, -7, 7, -7. Multiply by -1.

f $a_n = 80 \times (½)^n$

40, 20, 10, 5, 2½. Divide by 2.

6 The nth term of a sequence is given by $a_n = 3n + 1$.

a Put $a_n = 40$, find which term of the sequence 40 is

$40 = a_{13}$

b Find whether 100, 200 and 1000 are terms of the sequence

$100 = a_{33}$,
200 is not a term,
$1000 = a_{333}$

7 Answer each question by forming an equation and solving it.

a Find whether 44, 200 and 306 are terms of sequence $a_n = 10n - 6$

44 = a_5, 200 and 306 are not terms

b Find whether 40, 72 and 200 are terms of sequence $a_n = 2n^2$

40 is not a term, 72 = a_6, 200 = a_{10}

c Find whether 8, 96 and 128 are terms of sequence $a_n = 2^n$

8 = a_3, 96 is not a term,

128 = a_7

8 The nth term of a sequence is given by $a_n = 10n + 4$.

a Put $a_n < 100$, find how many terms are less than 100.

The 9 terms a_1 to a_9 are less than 100

b Put $a_n > 56$, find first term greater than 56. State its position number and value.

a_6 = 64

9 Answer each question by forming an inequality and solving it.

a How many terms of sequence $a_n = 2n - 5$ are less than 100?

52

b What is the first term of sequence $a_n = 7n - 44$ greater than 100?

a_{21} = 103

Series and Partial Sums

ⓘ Series

For a sequence a_1, a_2, a_3 ..., the **series** is the **infinite sum** of all the terms of a sequence.

$$a_1 + a_2 + a_3 + \cdots$$

ⓘ Partial Sum of a Series

The **partial sum** of a series is:

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n$$

Example Finding the partial sum of a series.

A series is $5 + 15 + 25 + 35 + 45$ ...

Find S_4

$S_4 = 5 + 15 + 25 + 35$
$= 80$

A series is $1 + 4 + 9 + 16 + 25 + 36$ ...

Find S_7

$S_7 = 1 + 4 + 9 + 16 + 25 + 36 + 49$
$= 140$

Practice

A series is $7 + 9 + 11 + 13 + 15$ ...

Find S_3

27

A series is $1 + 5 + 9 + 13 + 17$ ...

Find S_8

120

A series is given by $a_n = 2n - 1$

Find S_5

25

A series is given by $a_n = n^2 - 1$

Find $S_6 - S_2$

82

ⓘ Sigma Notation

We use sigma notation to write the partial sum of a sequence using its nth term formula:

$$\sum_{n=i}^{j} a_n = a_i + a_{i+1} + a_{i+2} + \cdots + a_{j-1} + a_j$$

This means starting from $n = i$ and ending at $n = j$, add up all the terms of the sequence a_n.

Interpret sigma notation.

Write out the sum of

$$\sum_{n=1}^{5} n^2 =$$

Write out the sum of

$$\sum_{n=3}^{6} (2n-1) =$$

Write out the sum of

$$\sum_{n=3}^{100} n^2 =$$

Write out the sum of

$$\sum_{n=10}^{75} 2n =$$

Write out the sum of

$$\sum_{n=4}^{7} (5n+1) =$$

Hence evaluate the sum.

114

Write out the sum of

$$\sum_{n=1}^{5} (-2)^n =$$

Hence evaluate the sum.

−22

1 Find the sum S_4 of the first four terms of each sequence.

a 3, 5, 7, 9, 11, 13, ...

24

b 2, 6, 18, 54, 162, 486, ...

80

c 2, 1, 1/2, 1/4, 1/8, ...

$3\frac{3}{4}$

d 32 - 16 + 8 - 4 + 2 - 1 + ⋯

20

2 Find the partial sums S_4, S_5 and S_6 for each series (which you will need to continue).

a 1 - 2 + 3 - 4 + ⋯

-2, 3, -3

b 81 + 27 + 9 + 3 + ⋯

120, 121, 121⅓

c 30 + 20 + 10 + ⋯

60, 50, 30

d 0.1 + 0.01 + 0.001 + 0.0001 + ⋯

0.1111, 0.11111, 0.111111

3 Rewrite each partial sum without sigma notation, then evaluate it.

a $\sum_{n=1}^{6} 2n$

42

b $\sum_{n=1}^{6} (3n+2)$

75

c $\sum_{k=3}^{7} (18-3n)$

15

d $\sum_{n=5}^{8} n^2$

174

e $\sum_{n=1}^{4} n^3$

100

f $\sum_{n=0}^{5} 2^n$

63

g $\sum_{n=2}^{4} 3^n$

117

h $\sum_{\ell=1}^{31} (-1)^{\ell}$

−1

i $\sum_{\ell=1}^{40} (-1)^{\ell-1}$

0

j $\sum_{n=5}^{105} 4$

404

k $\sum_{n=0}^{4} (-1)^n (n+5)$

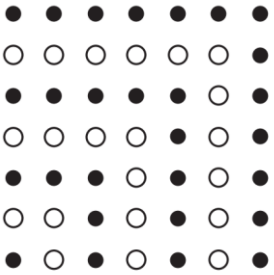
7

l $\sum_{n=0}^{4} (-1)^{(n+1)} (n+5)$

−7

4 Use the dot diagram to explain why the sum of the first n odd positive integers is n^2.

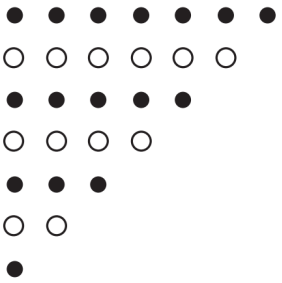
$$1 + 3 + 5 + 7 \ldots = n^2$$



Starting from the bottom left, adding each term extends the previous square by adding dots on two sides.

The diagram forms an n × n square.

5 Use the dot diagram to explain why the sum of the first n positive integers is $\frac{1}{2}n(n + 1)$.



$$1 + 2 + 3 + 4 \ldots + \frac{1}{2}n(n+1)$$

Starting from the bottom, each term adds height and length to the triangle.

The triangle can be paired with an identical copy, rotated 180°.

Together they form an n × (n + 1) rectangle.

The original triangle is half of this rectangle.

6 Rewrite each sum in sigma notation, starting each sum at $n = 1$. Do not evaluate it.

a

$1^3 + 2^3 + 3^3 + \cdots + 40^3$

$\sum_{n=1}^{40} n^3$

b

$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{40}$

$\sum_{n=1}^{40} \left(\frac{1}{n}\right)$

c

$3 + 4 + 5 + \cdots + 22$

$\sum_{n=1}^{20} (n+2)$

d

$2 + 2^2 + 2^3 + \cdots + 2^{12}$

$\sum_{n=1}^{12} 2^n$

e

$-1 + 2 - 3 + \cdots + 10$

$\sum_{n=1}^{10} (-1)^n n$

f

$1 - 2 + 3 - \cdots - 10$

$\sum_{n=1}^{10} (-1)^{(n+1)} n$ or $\sum_{n=1}^{10} (-1)^{(n-1)} n$